Republic of the Philippines

Laguna State Polytechnic University

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Ocular Checkpoint 4 Deliverables Report

Application integration and technical documentation draft

12 November 2026

Ocular is the current interface brand of GAZE. The repository and Python package retain the GAZE name. This document covers the six requested Phase 4 deliverables and provides a written report draft plus a companion presentation draft. It is prepared for the project team and instructor to review against the supplied hand-in requirements.

The application has integrated image, saved-video and optional camera processing, including saved-video seat and movement review. Technical integration evidence exists. CP4 held-out classification results, representative demo-laptop throughput and the effort-weighted 80 percent completion calculation are not yet evidenced. They remain explicit submission gaps. [S01–S04]

Deliverable	Evidence in this draft	Status
1 Working application	Implemented paths and integration boundary	Technical evidence
2 Interface and outputs	Input modes, review workflow and exported records	Technical evidence
3 Metrics and laptop speed	Historical CP3 metrics and CP4 evaluation protocol	CP4 results pending
4 Analysis and limits	Recorded failures and operating constraints	Representative checks pending
5 Test cases	20 recorded cases with expected and actual outcomes	Historical execution
6 Report and slides	Written technical sections and 12 slide draft	Draft for team review

Source snapshot: d60038d on feat/ocular-rebrand. Historical tests and results retain their original dates and hardware. No new model evaluation, recording, submission, commit or tag is implied by this document. Live GitHub Issue 30 and Project state could not be retrieved; no live task status is claimed.

1 Working application and integration

The current implementation offers three input paths through the local Next.js interface and shared Python modules. The source code separates reusable detection, preprocessing, direction classification, seat association, histories and movement logic under src/gaze. Scripts coordinate processing rather than duplicating the algorithms. [S02, S14, S17]

Processing path	Implemented behavior	Evidence boundary
Image and automatic saved video	Uploads call scripts/run.py using the head model or geometric baseline. Annotated media and direction records are saved.	Automatic directions are camera-relative. Counts are observations, not identities.
Saved video seat review	Seat suggestions/manual regions, configurable movement rules, synchronized overlays, event seeking, histories, downloads, Stop and reopening.	Suggestions need correction. Real seat consistency and event accuracy are unmeasured.
Optional browser camera	Device/method selection, Start/Stop, elapsed capture time and matching processed-frame overlays. Frames stay in memory.	Direction-only. Physical-camera and representative three-seat acceptance remain pending.

Integration evidence

The saved-review verification record reports 157 application tests, 16 automation tests and 26 frontend tests after the faculty setup refinement, for 199 technical tests. It also records real-model API processing, browser workflow checks, cancellation and reopening. These are historical checks from 29 October 2026, not new executions on this document date. [S04]

On a 24-frame, 384 by 288 smoke clip built from a permitted Pointing04 image, the baseline produced 24 forward observations and no events. The head model produced 24 right observations and one event. This difference verifies that the pipeline and UI expose method outputs; without independent clip labels, it does not determine which method was correct. [S04, S10]

Completion assessment

The project has substantial integrated behavior, but no agreed effort weights or accepted three-seat laptop demonstration are recorded. Overall completion must use 100 times accepted completed effort divided by total planned effort. No 80 percent implementation claim is made from the number of screens, passing tests, or the checkpoint name. [S03]

The remaining acceptance work is a representative authorized multi-seat demonstration, another member’s setup reproduction, held-out scoring, laptop performance measurement and team review. These gaps affect completion even where modules and interface controls already exist. [S03, S04]

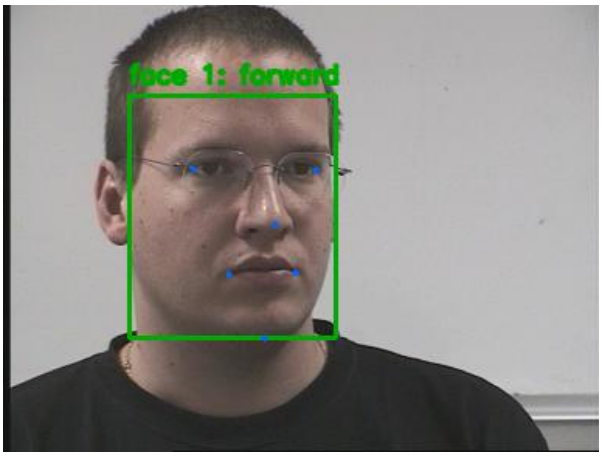
2 Interface and visualization of results

The website workspace at /analyze provides Image, Saved video and Camera modes. Saved video offers the automatic direction-only path and a seat-and-movement review path. The faculty workflow presents seat suggestions and manual correction before Confirm seats and process, with numerical coordinates and advanced thresholds collapsed by default. [S02–S04, S17]

Stage	Visible controls and outputs
Input and setup	Media preview, method choice, optional mirrored-input setting, fixed seat suggestions, rename/redraw/remove/add controls and plain-language active rules.
Processing	Processing state and Stop, shared inference reservation, input validation and clear failures. No fabricated percentage or instantaneous speed claim.
Review	Source-time playback, seat regions, detection boxes, direction labels, missing/ambiguous states, filterable movement records and event seeking.
Records	Per-seat histories, settings and source hash, observations, events, summaries, compact review JSON, annotated AVI and completed-session reopening.

Output meaning

Forward, left, right and down describe visible head orientation. Unknown indicates no usable direction; unassigned indicates that a detection cannot be reliably associated with a configured position. Seat IDs identify places rather than students. Detector scores, where retained, are not calibrated confidence in head direction or misconduct. Boxes, labels and counts satisfy the relevant visualization examples; masks and confidence percentages are not invented. [S01, S13, S18]



Historical permitted Pointing04 baseline output. This demonstrates annotation style, not a current Ocular screenshot or CP4 accuracy result. Source: N. Gourier, D. Hall and J. L. Crowley. [S07]

The recorded browser checks cover the integrated workflow and a 390-pixel viewport. They do not replace a human keyboard/usability check on the actual demonstration laptop. Camera review remains an optional separate mode without seat histories, event logging or recordings. [S03, S04]

3 Evaluation results and comparison with CP3

No CP4 held-out evaluation artifact or representative demo-laptop benchmark was found in the inspected source snapshot. The table below preserves the CP3 fixed stress-set results. It must not be relabeled as a new CP4 measurement. [S03, S05, S06]

CP3 method	Correct	Accuracy	Macro F1	Coverage	ms per image
Geometric baseline	29/63	46.03%	0.4875	52.38%	7.66
Learned head model	13/63	20.63%	0.3071	38.10%	83.48

CP3 used an Apple M5 MacBook Air with CPU inference on 63 images. The reported baseline/main rates are 130.54 and 11.98 images per second. Their timing includes image decoding, preprocessing, inference and record construction; it excludes model initialization, checksum preflight, annotation/report writing and an explicit warmup exclusion. Image rates are not end-to-end saved-video FPS. [S05, S06]

Comparison status

Measure	CP3 record	CP4 record and change
Held-out accuracy and macro F1	46.03% and 0.4875 baseline; 20.63% and 0.3071 main	Not measured for CP4; numerical change unavailable.
Coverage	52.38% baseline; 38.10% main	Not measured for CP4; numerical change unavailable.
Integrated laptop video speed	No equivalent integrated workflow timing in this comparison	Not measured on the demo laptop.
Application workflow	Automatic image/video results; separate configured seat processing	Web seat setup, source-time event review, cancellation and reopening now integrated.

Historical speed evidence

The CP2 FourPeople baseline benchmark recorded a median 10.95 processing FPS on an AMD Ryzen 5 5600X desktop at 1280 by 720, with four participants and tile grid 2. Three runs measured 11.18, 10.95 and 10.41 FPS after five warmup frames. This is a desktop proxy measurement on an older implementation, not the CP4 demo-laptop result. [S11]

On the CP3 unmodified subset, both methods were correct on 8 of 15 images. For six strong unmodified portrait profiles, the head method was correct on 6 of 6 versus 1 of 6 for the baseline. This narrow success does not offset the main method’s worse aggregate stress-set result. [S05, S09]

3 Evaluation protocol still required for CP4

Held out set and evaluator

The supplied Phase 4 instructions require held-out testing. This supersedes the older CP5-only sequencing, but the revised frozen protocol and resulting score file are not present. The existing Pointing04 manifest reserves 120 images from subjects 11 to 15, separated from 240 validation images from subjects 1 to 10. The CP3 stress set has already been inspected and evaluated, so it cannot serve as a newly untouched test set for improvements selected after CP3. [S03, S07, S08]

The required src/evaluate.py entry point is absent. scripts/evaluate.py currently accepts only the validation split and baseline/approximate pose modes; scripts/evaluate_cp3.py is the recorded CP3 evaluator. src/gaze/evaluation.py contains reusable metric calculations, but is not the requested executable entry point. A CP4 evaluator must be implemented or the instructor must accept a documented alternative location before the deliverable is claimed. [S03, S15, S16]

Metric definitions

Metric	Definition and reporting rule
Accuracy	Correct predictions divided by all eligible labeled images. Missing detections, unknowns and ambiguous cases count against the denominator.
Macro F1	Unweighted mean of the F1 scores for forward, left, right and down. Report per-class support and precision/recall.
Coverage	Usable direction predictions divided by all eligible labeled images. Report unavailable counts and reasons separately.
Confusion matrix	Ground-truth rows and prediction columns, including an unknown prediction column.
Video performance	Seat consistency and event precision/recall require independent multi-person/temporal labels; writing false alerts require measured writing exposure.

Execution and laptop measurement plan

1. Verify permissions, human label checks, class counts and source/subject separation. Freeze the manifest, preprocessing, model hashes, thresholds and implementation revision before test inference.

2. Score both methods on identical held-out inputs with identical label mapping and failure accounting. Save machine-readable results, per-sample outcomes and confusion matrices without tuning on the test errors.

3. Run the representative saved-video review workload on the actual demo laptop with at least three visible fixed seats. Record CPU, OS, runtime versions, power conditions, source size/rate, face count, settings and model versions.

4. Repeat the same workload three times with a declared warmup. Report median processing FPS and the exact included/excluded stages. Keep source-video time distinct from wall-clock processing time.

5. Report CP4 minus CP3 accuracy in percentage points and the F1/coverage changes only where inputs and conditions permit a meaningful comparison. Explain dataset or hardware changes beside any cross-checkpoint numbers.

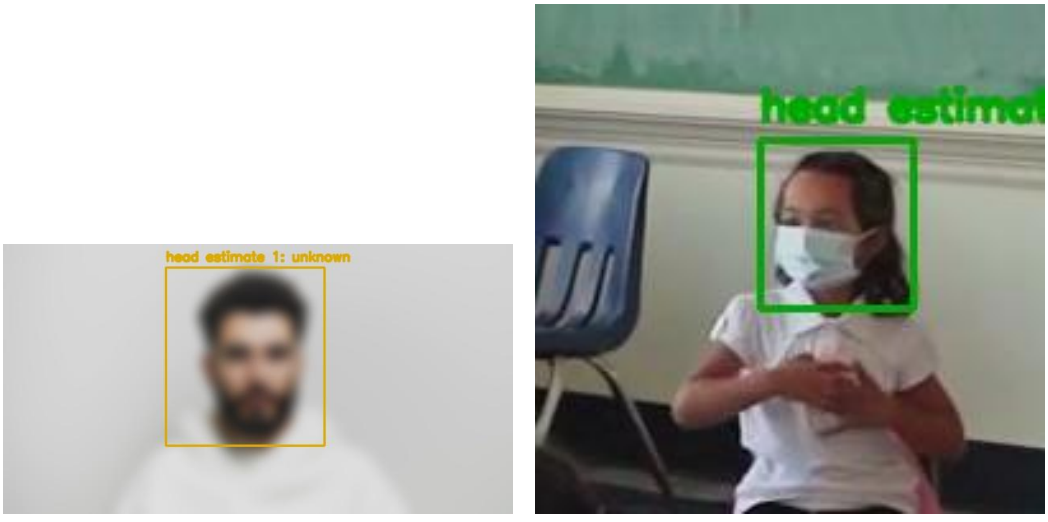
4 Performance analysis from recorded evidence

The CP3 failure groups provide measured examples of limits. A failed sample is an incorrect or unavailable prediction. The conditions below label the inputs; they are not controlled causal proof. Counts include all 63 samples rather than only the selected gallery. [S09]

Input condition	Baseline failures	Main failures
Blur	9	12
Clean pose	0	5
Low light	6	12
Mask and classroom clutter	2	2
Occlusion	6	10
Profile angle	5	0
Small image	6	9
All conditions	34	50

Of the main method's 39 unavailable predictions, 21 had no detection, 13 failed the blur quality gate and five fell outside the supported pose range. These recorded reasons explain why coverage falls; they do not identify every physical cause of a miss. Two masked classroom crops were usable but predicted forward when the reference label was right. [S06, S09]

Examples of observed failures



Recorded CP3 examples: left, forward becomes unknown with head_blurry; right, a reference right label is predicted forward. Pexels sources: cottonbro studio and RDNE Stock project. These examples describe head pose only. [S08, S09]

The baseline depends on usable facial landmarks and had poor coverage for strong profiles. The learned model handled the six strong portrait profiles well, but lost more samples to detection/quality limits in the overall stress set. Improvements should be developed on distinct development data and checked on a frozen held-out set. [S05, S08, S09]

4 Operating conditions and remaining limitations

Area	Supported or demonstrated condition	Limit or unvalidated condition
Capture layout	One fixed camera and fixed seat regions; manual correction of suggested regions.	Moving cameras, seat changes and unrestricted classroom coverage are outside scope.
Input compatibility	Saved review accepts compatible constant-cadence MP4/WebM and checks decoded timestamps.	Unavailable, duplicate, nonfinite or irregular timestamps fail compatibility checks.
Resource bounds	Review limits: 64 MB upload, five minutes, 18,000 frames, 1920 by 1080 pixels, 20 seats.	64 MB compact review JSON and 512 MB output budget; these limits are not throughput guarantees.
Observation quality	Clear sufficiently visible heads can produce direction observations; unknown reasons are retained.	Small, blurry, dim, occluded, mixed or unsupported views can be unknown or wrong.
Local operation	A local server coordinates upload, review and optional camera processing.	No direct CCTV/RTSP/ONVIF connection, public multi-user service or identity recognition is claimed.
Human interpretation	Events organize sustained/repeated movements for an instructor to inspect.	Intent, eye gaze, cheating detection and disciplinary decisions are not inferred.

Limits observed during integration

The FourPeople proxy's 601 frames passed timing inspection with maximum cadence deviation of about 0.333 milliseconds. After correction of a nearby-head padding conflict, suggestions returned five regions for four participants, including an extra small candidate that needed manual removal. This is evidence for faculty confirmation, not perfect seat detection. [S04]

Historical movement diagnostics changed from 12 repeated records to one after adding minimum episode dwell, while observation counts stayed unchanged. The result demonstrates filtering behavior. It does not establish event precision, recall or a reduction in verified writing false alerts. [S11]

Single-person Pointing04 poses and correlated CP3 transformations do not establish performance for distant or crowded classrooms. Independent labels for multi-seat association and real event intervals, ordinary-writing exposure and the target laptop workload remain required. [S03, S07, S08]

Storage and failure behavior

Saved review keeps source media and provenance in local ignored run directories. Cancellation and failed runs do not publish completed-session manifests; partial local outputs can remain for manual cleanup. Source access and retention must be agreed before collecting classroom footage. [S04]

5 Recorded technical test cases

Execution record: 29 October 2026. All outcomes below are transcribed from the repository test-case record, not rerun for this draft. Pass means the named technical expectation was met. It does not certify classroom direction accuracy, real event accuracy or operation on the demonstration laptop. Cases TC05 to TC11 and TC13 to TC20 include hard or edge conditions. [S10]

ID	Input	Condition	Expected	Actual	Recorded result
TC01	24-frame sample WebM; baseline	Valid seat/settings	Process all frames; persist settings/history/events with source hash.	24 frames; 24 forward observations; zero movement events; matching exports/hash.	Pass
TC02	Same WebM; automatic head model	Same input and seat	Process frames and expose movement outputs without assuming accuracy.	24 frames; 24 right observations; one movement event.	Pass
TC03	Browser upload of same WebM	Visual seat drawing plus numeric edit/rename	Save the edited region/name and selected duration.	Renamed desk-A persisted; real head inference completed and event displayed.	Pass
TC04	Completed real-model session	Select movement-event start	Pause/seek to event source time; show matched annotations.	Browser seek matched start within one source frame; source-sized overlay and provenance/history shown.	Pass
TC05	Saved synthetic event fixture	Edge: missing frame at 1.5 s	Show unknown/missing; do not bridge continuous movement or retain flag.	Player showed missing; Python gap tests split intervals and events.	Pass
TC06	Synthetic 10-frame sequence	Edge: two detections in one region	Mark seat ambiguous/unassigned; reset history/event continuity.	Ambiguous/unknown recorded, active review flags cleared, later event restarted at 1.5 s.	Pass
TC07	Overlapping/duplicate/zero-area/outside seats	Edge: invalid configuration	Reject before inference/session publication.	Python and frontend validators rejected cases; overlapping-seat API returned 400.	Pass
TC08	Nonempty corrupt WebM	Edge: unreadable media	Clear failure; no completed review record.	API returned 422; Python test confirmed no review.json publication.	Pass
TC09	Decoded timestamps [0,17,80,97] at 60 FPS; unavailable/duplicate timestamp variants	Edge: irregular or unavailable timing	Automatically reject timing incompatible with frame-indexed review.	Timing checker rejected variable, duplicate, missing and nonfinite timestamps; no user-attestation checkbox is required.	Pass
TC10	Saved source HTTP requests	Edge: valid/out-of-bounds byte ranges	Valid range returns 206 and exact bytes; invalid range returns 416.	Requested first 32 bytes returned 32 bytes; out-of-range returned 416.	Pass

ID	Input	Condition	Expected	Actual	Recorded result
TC11	Artifact URL and foreign browser origin	Edge: disallowed access	Reject unknown/prototype asset and nonlocal origin.	__proto__ artifact returned 404; foreign Origin returned 403.	Pass
TC12	Processing FourPeople proxy	Stop during processing	Abort child, release inference, do not add completed session.	Completed-session count unchanged; subsequent camera start/stop succeeded.	Pass
TC13	Final manifest publication	Edge: abort during actual rename	Remove completed manifest and report cancellation.	Filesystem regression test aborted after rename; session.json was absent.	Pass
TC14	Saved-session bookmark after reload	Edge: session omitted from recent list	Reopen intact session directly without inference.	Browser test hid list entries; metadata fallback reopened saved URL.	Pass
TC15	Times around 0.25 s at 4 FPS	Edge: exact frame boundary/outside coverage	Use half-open frame intervals; return no frame past processed duration.	0.249→frame 0, 0.25→frame 1, 0.75→no frame for 0.75 s coverage.	Pass
TC16	Seat mirror round trip	Already mirrored pixels	Convert drawn/canonical rectangles once in each direction.	Coordinate assertions round-tripped within 1e-10.	Pass
TC17	Invalid movement values/output limits	Edge: NaN/zero/fractional count/limit breach	Reject invalid settings or over-budget session without completed record.	Movement validators and simulated frame/output-budget boundary tests passed.	Pass
TC18	390px browser viewport; existing modes	Responsive navigation/regression	No horizontal overflow; preserve basic video/image workflows.	No overflow; Directions only and Image controls reachable; no page JavaScript errors.	Pass
TC19	Repeated nearby synthetic head boxes across three frames	Edge: close seats and padding conflict	Keep distinct raw detections separate without overlapping regions.	Regression tests retain both seats; padding shrinks before any raw-region conflict is rejected.	Pass
TC20	Empty detector outputs across three frames	Edge: suggestions miss every head	Return no invented seats and retain manual fallback.	Suggestion unit test returns empty seats and manual-setup guidance.	Pass

The baseline and main smoke tests use a repeated permitted Pointing04 image and test integration only. Synthetic missing/ambiguous/event fixtures test mechanics separately. The recorded suite includes 20 cases, with at least three hard cases. The remaining action is representative human execution and labeled video correctness cases on the actual demo laptop. [S04, S10]

6 Technical report draft

Introduction

Instructors supervising examinations may need to revisit when visible head movements occurred and how long they remained observable. Brief turns and looking down during ordinary writing can have similar visual patterns, so a movement alone cannot establish misconduct. Ocular, previously presented as GAZE, is a local computer vision review aid that organizes these observations while preserving the instructor's role in interpretation. [S01, S02]

The project aims to detect visible faces or heads, classify forward/left/right/down orientations, associate observations with fixed seat positions, maintain source-time histories, and export sustained or repeated movement records. Its main demonstration uses saved video from one fixed camera. The optional camera mode provides direction observations without seat histories or recordings. The work excludes identity recognition, eye-gaze or intent estimation and automatic disciplinary decisions. [S01, S02]

The current checkpoint focuses on integration of processing, faculty setup and review. The interface now connects fixed-seat setup, timestamped outputs, movement records, downloads and saved-session reopening. Technical checks support the integration claims, while real classroom effectiveness and the required CP4 held-out/laptop measurements remain separate questions. [S03, S04, S12]

Data

The initial direction dataset uses the author-hosted Pointing04 database with its recorded attribution condition. The repository data card documents 15 subjects and nominal pose targets in 384 by 288 color images. The selected manifest has 240 validation images from subjects 1 to 10 and 120 locked test images from subjects 11 to 15, with 60 and 30 examples per direction respectively. Twenty committed examples come from validation. Source and subject separation prevent reuse of a person or recording across these splits. No local model training is performed. [S07]

CP3 used a separate fixed stress set from three Pexels recordings by cottonbro studio, Nicola Barts and RDNE Stock project. The set contains 15 original frames and 48 deterministic blur, dimming, occlusion and size variants, for 63 images. Its forward/left/right/down counts are 16/15/17/15. Labels were assigned before inference, with a human cross-check still pending. The small number of people and correlated transformations limit generalization and statistical independence. [S08]

The public-domain FourPeople office-meeting clip supports multi-person integration and historical speed diagnostics. It is a proxy, not classroom event ground truth. The 24-frame integration smoke clip and synthetic event fixtures are explicitly separate from evaluation sets. Neither should enter held-out accuracy denominators. For CP4, the team must finalize a permitted, independently checked held-out protocol and representative labeled video before reporting new correctness results. [S03, S04, S10, S11]

6 Method and experiments draft

Method

The baseline uses MediaPipe facial landmarks with shared preprocessing and eye–nose–chin geometry to classify direction. The main automatic method combines Gold-YOLO head detection with 6DRepNet360 rotation estimates using CPU ONNX Runtime. It can estimate sufficiently visible profile or rear views without facial landmarks, but cannot recover information from full occlusion. Quality checks retain unusable, mixed or unsupported observations as unknown. The approximate six-point PnP comparison remains available through the documented Python paths. [S02, S14, S18]

Automatic head labels use camera-facing neutral axes with subject-relative left/right. The web seat-review path attaches those observations to fixed image positions; assigning a seat does not calibrate the head direction to a desk. A separate configured head-processing mode uses manually selected forward references and source/model/seat-bound neutral calibration. These direction bases must remain distinct in interpretation and evaluation. [S04, S12, S18]

Each configured region accepts one reliable observation. Missing or multiple competing detections produce missing or ambiguous states. Histories use source timestamps, retain explicit unknown entries and reset continuity across gaps. Sustained/repetition rules use duration, repetition count/window, minimum episode dwell and maximum gap settings. JSON records retain the source hash, method and settings so the instructor can trace each output. [S13]

The Next.js interface calls local Python commands and displays annotated outputs. Saved-review playback synchronizes source time with boxes, seat labels, histories and descriptive movement records. Selecting an event seeks to its source timestamp; completed sessions reopen without rerunning inference. Shared processor reservations and cancellation bound overlapping work. The camera extension keeps frames in memory and releases capture/processing resources on Stop and lifecycle failure. [S02, S04, S17]

Experiments

The recorded CP3 experiment froze model/settings and input fingerprints before scoring identical prepared pixels with both methods. Accuracy counts unknowns and missed detections as incorrect; macro F1 averages the four direction classes; coverage reports the fraction with usable direction predictions. The baseline achieved 29/63 correct, macro F1 0.4875 and 52.38 percent coverage. The head method achieved 13/63 correct, macro F1 0.3071 and 38.10 percent coverage. The results do not show aggregate improvement. [S05, S06, S16]

CP4 integration experiments recorded real-model smoke processing, event-gap fixtures, malformed inputs, cancellation, downloads, synchronized seeking and session reopening. Their purpose is to verify the implemented workflow, not estimate real-world event precision or held-out direction accuracy. No new CP4 aggregate score or representative demo-laptop FPS is reported in this draft. [S03, S04, S10]

The pending CP4 experiment must freeze a revised protocol, provide the requested evaluator entry point or an approved equivalent, and score baseline/main methods on the same untouched inputs. Laptop measurements must use the integrated saved-video workload with documented settings, included stages, warmup and repeat runs. A valid comparison with CP3 must state whether data, preprocessing and hardware changed. Test-error inspection may guide later development but must not be used to tune the held-out evaluation being reported. [S03]

6 Draft presentation and remaining hand in evidence

The companion draft presentation contains 12 editable slides. The sequence below is also a presentation script outline, with references and qualifications in the slide notes. Replace pending measurements only after the corresponding evidence exists.

Slide	Subject	Presentation content
1	Ocular Checkpoint 4	Project identity, local review-aid purpose and draft status.
2	Problem and scope	Instructor review, one fixed camera, direction observations and human interpretation.
3	Data and separation	Pointing04 validation/test split, CP3 stress set and separate video/technical fixtures.
4	Application integration	Next.js inputs and shared Python paths, seat setup, histories, events and reopening.
5	Baseline and main methods	Facial-landmark geometry, head detector plus rotation model and direction-basis distinction.
6	Interface demonstration	Upload, confirm seats, process, inspect records, seek and reopen a saved session.
7	Recorded CP3 results	Historical accuracy/F1/coverage and the image timing scope.
8	CP4 evaluation gap	Held-out evaluator, frozen protocol and representative laptop benchmark still required.
9	Technical test cases	20 recorded cases, hard inputs and the limited meaning of technical pass results.
10	Observed failures	Blur, missed detections, clutter/occlusion and the extra seat suggestion.
11	Operating limitations	Fixed camera/seat assumptions, bounded media, explicit gaps and unvalidated real-event quality.
12	Remaining submission evidence	Metrics, human/laptop checks, draft review and the separate recording/tag/submission gate.

Items still required before a complete submission

Deliverable 3 remains incomplete until held-out metrics and demo-laptop speed exist. The team must also verify the representative three-seat workflow, review the effort-weighted completion claim, and confirm the accuracy and interpretation of the draft. This document and deck do not close those gates. [S03]

Deliverable 7, the backup recording of five minutes or less, is outside this document’s requested scope and has not been created. The team should tag cp4-80 only on the reviewed actual hand-in commit after checking all submission evidence and authorizing publication. No commit, tag or remote submission is performed by this document task. [S03]

Evidence references and provenance

Bracketed references point to repository records inspected at source revision d60038d. Links below are pinned to that revision, so historical measurements and technical test results remain attributable. The draft itself is dated 12 November 2026; the cited experiments retain their original dates.

[S01] Product requirements and supported behavior. PRD.md

[S02] Current application architecture and processing paths. ARCHITECTURE.md

[S03] Phase 4 requirements and outstanding evidence. docs/checkpoints/cp4-checklist.md

[S04] Saved review verification and faculty setup refinement. docs/superpowers/specs/2026-10-29-saved-review-verification.md

[S05] CP3 hand in index and hardware description. docs/checkpoints/cp3-hand-in/README.md

[S06] CP3 complete metrics and measurement provenance. outputs/evidence/cp3-core/comparison.json

[S07] Pointing04 source permissions and split data card. docs/evaluation/data-card.md

[S08] CP3 source permissions labels and stress set data card. docs/checkpoints/cp3-hand-in/data-card.md

[S09] CP3 failure groups and selected output gallery. outputs/evidence/cp3-core/README.md

[S10] Recorded CP4 test cases. docs/checkpoints/cp4-test-cases.md

[S11] Historical video throughput and movement rule diagnostics. docs/evaluation/video.md

[S12] Saved video review before and after. docs/checkpoints/cp4-before-after.md

[S13] Fixed seat and observation gap semantics. docs/evaluation/seats.md

[S14] Automatic image and video processing entry point. scripts/run.py

[S15] Current validation only evaluation entry point. scripts/evaluate.py

[S16] Metric definitions and missed observation accounting. src/gaze/evaluation.py

[S17] Current website setup and usage. frontend/README.md

[S18] Experimental calibrated head model details. docs/evaluation/hidden-faces.md

The original university seal is reused from docs/checkpoints/assets/lspu-seal.png, extracted from the supplied CP2 report. Its presence is a formatting choice and does not establish instructor or university acceptance.

Source limitations: local GitHub task metadata is a snapshot. Issue 30 could not be retrieved during preparation, and the private Project state was not verified. No assignee, Project status, academic acceptance or hand-in publication is asserted.